

Appendix: Alternative Density Estimators

This appendix describes six additional density estimators and reports reduced figure panels for these supplementary methods across the running examples (Sections 3.1.1–3.2.2); the main-text KDE and IWMDE curves are not repeated. Two considerations apply when selecting the density estimators. First, only KDE, IWMDE, CMDE, and the truncated bivariate normal directly extend to multivariate γ ; the logspline, log-concave MLE, and Bayesian nonparametric estimators are effectively restricted to the univariate case. Second, the model-based estimators (CMDE, IWMDE) exploit the known prior–parameter relationship and achieve near-exact accuracy at modest sample sizes, whereas estimators that treat the marginal posterior of γ as an opaque distribution require either large samples (nonparametric) or distributional assumptions (parametric).

Figures 6–9 report reduced appendix displays for the supplementary density estimators omitted from the main-text Figures, and Table 1 reports accuracy for KDE and IWMDE as a function of the number of posterior draws in the univariate t -test.

4.1 Logspline Density Estimation

Logspline density estimation (Kooperberg & Stone, 1991) models the log-density as a polynomial spline with BIC-selected knots, respecting known support boundaries. We use the `logspline` package (Kooperberg, 2020) with explicit lower and upper bounds matching the support of the hyper-prior. The estimator is well suited to the smooth unimodal posteriors of γ encountered here, but is limited to the univariate case.

4.2 Log-concave Maximum Likelihood Estimation

The log-concave MLE (Dümbgen & Rufibach, 2009) computes the nonparametric density whose log is concave, requiring neither bandwidth nor knot selection. We use the smoothed variant from the `logcondens` package (Rufibach & Dümbgen, 2023). The concavity constraint rules out multimodality and heavy tails, and the smoothed estimator is computationally expensive for large samples.

4.3 Bayesian Nonparametric Density Estimation (BNPmix)

As a fully Bayesian alternative, we fit a Dirichlet-process mixture to the posterior samples of γ via the `PYdensity` function of the `BNPmix` package (Corradin et al., 2021), with a Pitman–Yor prior (strength 1, discount 0, reducing to a Dirichlet process), 5,000 iterations, and 2,000 burn-in. This approach returns a full posterior distribution over density evaluations, allowing uncertainty in the density estimate to be propagated into the Bayes factor sensitivity curve, but it is orders of magnitude slower than the other alternatives and offers no practical benefit for the smooth unimodal posteriors encountered in our examples.

4.4 Truncated Normal Approximation

The simplest parametric alternative fits a normal distribution truncated to the support of γ by maximum likelihood, with the usual truncation correction $Z = \Phi((\gamma_U - \mu)/\sigma) - \Phi((\gamma_L - \mu)/\sigma)$. The two-parameter family is adequate when the posterior of γ is approximately symmetric and bell-shaped, but fails to capture skewness and can produce badly biased density ratios when the true posterior deviates from normality.

4.5 Truncated Asymmetric Power Distribution (APD)

A more flexible four-parameter family truncates the asymmetric power distribution of Komunjer (2007) to the support of γ . The parameters (location, scale, asymmetry $\alpha \in (0, 1)$, shape $\delta > 0$) are estimated by maximum likelihood with a numerical normalizing constant obtained by integration over the truncated support. The APD accommodates skewness and variable tail weight, but the four-parameter optimization is slow and occasionally fails to converge.

4.6 Conditional Marginal Density Estimator (CMDE)

The CMDE (Gelfand & Smith, 1990; Chen, 1994), also known as the Rao–Blackwell density estimator, is the variance-minimizing special case of the IWMDE obtained when the weighting function equals the full conditional posterior of γ . It averages this exact conditional over the posterior draws of the remaining parameters:

$$\hat{p}(\gamma^* \mid y, \mathcal{H}_\gamma) = \frac{1}{n} \sum_{i=1}^n p(\gamma^* \mid \theta_i, y, \mathcal{H}_\gamma). \quad (14)$$

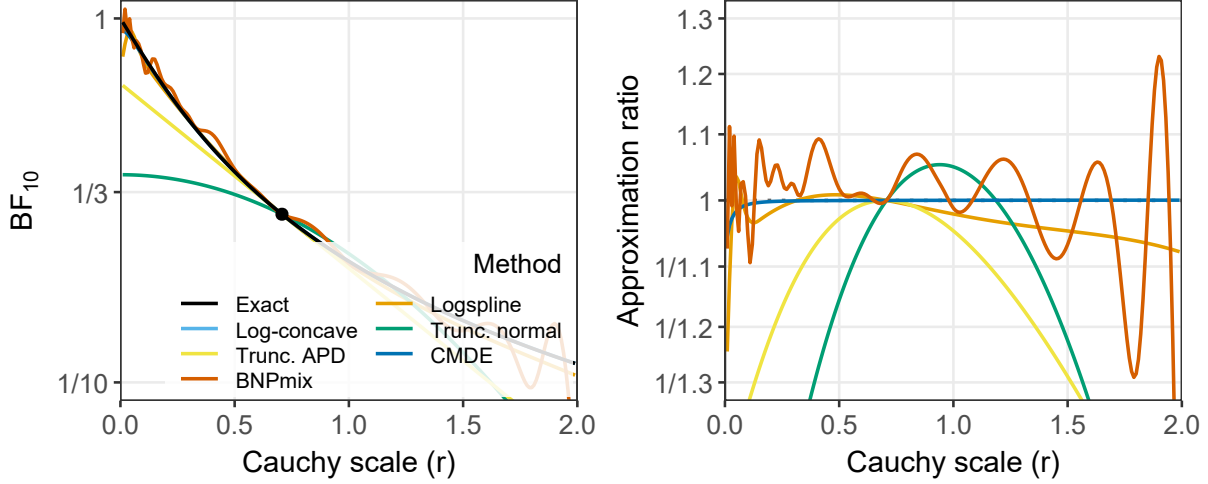


Figure 6: Univariate sensitivity analysis for the Bayesian t -test: reduced appendix display of the supplementary density estimators. Left panel: Bayes factor BF_{10} as a function of the Cauchy scale parameter r . Right panel: approximation ratio BF_{approx}/BF_{exact} . The black solid line shows the exact solution; the colored lines show logspline density estimation, the log-concave maximum likelihood estimator, the truncated normal approximation, the truncated asymmetric power distribution (APD), the conditional marginal density estimator (CMDE), and the Bayesian nonparametric mixture estimator from BNPmix.

For the Cauchy-prior t -test, the full conditional is $p(\gamma \mid \delta) \propto \gamma/(\delta^2 + \gamma^2)$ with normalizing constant $(1/2) \log(1 + 4/\delta^2)$ on $\gamma \in [0, 2]$. The CMDE is asymptotically optimal but requires a closed-form full conditional that is analytically normalizable, a condition rarely satisfied outside simple models.

Table 1: Computation time (seconds) and approximation accuracy by number of posterior draws and density estimation method. The two methods are the kernel density estimator (KDE) and the importance-weighted marginal density estimator (IWMDE). Mean absolute error (MAE) and root mean squared error (RMSE) are computed on the log scale of the approximation ratio, $\log(BF_{approx}/BF_{exact})$. Subscript t denotes truncated measures that exclude the outer 5% of the support on each side.

n	KDE				IWMDE			
	MAE	RMSE	MAE _t	RMSE _t	MAE	RMSE	MAE _t	RMSE _t
3,000	0.1820	0.2179	0.1583	0.1727	0.0008	0.0037	0.0003	0.0003
10,000	0.0729	0.1413	0.0502	0.0688	0.0012	0.0027	0.0008	0.0009
30,000	0.0495	0.1121	0.0318	0.0402	0.0020	0.0046	0.0011	0.0016
100,000	0.0337	0.0790	0.0219	0.0288	0.0022	0.0045	0.0014	0.0022
300,000	0.0313	0.0649	0.0244	0.0292	0.0011	0.0023	0.0008	0.0011

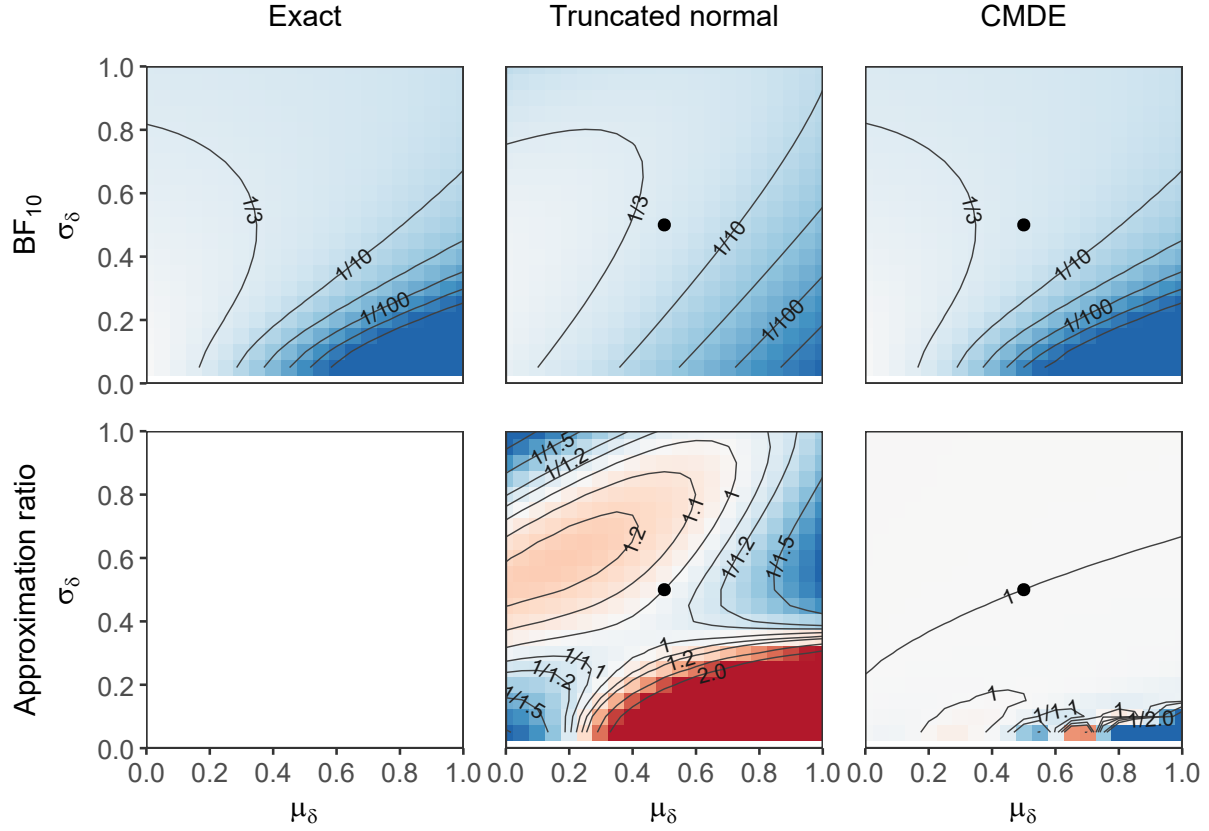


Figure 7: Bivariate sensitivity analysis for the informed Bayesian t -test: reduced appendix display of the supplementary methods. Columns correspond to the exact computation, the truncated bivariate normal approximation, and the conditional marginal density estimator (CMDE). Top row: Bayes factor BF_{10} as a function of the prior mean μ_δ and prior standard deviation σ_δ ; contour labels show BF_{10} levels (darker colors indicate more extreme evidence), and the filled circles mark the anchor at $(\mu_\delta, \sigma_\delta) = (0.5, 0.5)$ in the approximation panels. Bottom row: approximation ratio $BF_{\text{approx}}/BF_{\text{exact}}$ (darker colors indicate more extreme ratio, ratios larger than one in red, ratios lower than one in blue); the lower-left panel is blank because the exact computation has no approximation ratio.